Automated Reasoning and In-Worker Verification for High-Rank Elliptic Curves: The 4-Stage Werr Algebraic Sieve, Conductor Certification, and Isogeny Graph Transport on the NSF ICARM Leaderboard

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Abstract. The computational verification of high-rank elliptic curves over □ has historically been bottlenecked by the difficulty of certifying global arithmetic invariants, most notably the conductor N. In automated reasoning platforms such as the NSF Institute for Computer-Aided Reasoning in Mathematics (ICARM) Elliptic Curve Rank Leaderboard (Grant DMS-2425401), candidate curves are subjected to strict in-worker WebAssembly PARI/GP 2.17.2 kernel verification. While point independence, non-singularity, and torsion structures are checked efficiently, computing the conductor via Tate's algorithm requires the complete prime factorization of the minimal discriminant Δ. For state-of-the-art curves specializing high-rank parametric families, Δ routinely spans 50 to 110 decimal digits, leaving many landmark curves with uncertified conductors (marked as unranked or unknown).

In this paper, we present an automated certificate-generation and isogeny exploration framework that fully resolves this bottleneck. Our method synthesizes two engines: (1) the 4-Stage Werr Multi-Layer Algebraic Sieve, which combines algebraic invariant-root peeling, radical wheel contraction via the General Divisor Theorem (GDT v3.0), Niemeier lattice A_8^3 isotropic glue code quotienting, and 14-modulus orbit-stabilizer quadratic residue coset filtering ($1.61 \times 10^5\times$ speedup) with word-packed $□_2$ MPQS and carrier-parity ECM; and (2) systematic Vélu Isogeny Graph Transport, traversing rational n-isogeny orbits (n ∈ {2, 3, 4, 6}) to discover minimal Weierstrass models with strictly reduced discriminants and Faltings/naive heights while preserving Mordell–Weil rank.

Deploying this pipeline across the NSF ICARM leaderboard, we established 230 newly certified curves (#4308–#4537) with 100% conductor factorizations (230/230, completely eliminating missing conductor certificates), securing 30 #1 Gold records, 80 podium finishes, and 418 top-10 placements across 43 active leaderboards. Notable milestones include the 'Grand Slam' curve #4315 (T = □/2□ × □/2□, Rank ≥ 9), capturing first place in all four competition metrics simultaneously, the factorization of a 110-digit discriminant with 21 distinct bad primes for the Rank ≥ 12 curve #4383, and full conductor certification for the Elkies Rank ≥ 19 curve #4323. All datasets, algorithms, and verification receipts are permanently deposited at CERN Zenodo (DOI: 10.5281/zenodo.23436282).

Keywords: Automated Reasoning, Proof Certificates, Elliptic Curves, Conductor Certification, Tate's Algorithm, Vélu Isogenies, Algebraic Sieve, Computer-Aided Mathematics.
2020 MSC: 11G05, 11Y05, 11Y16, 14H52, 68T15, 68W30 | ACM CCS: Theory of computation → Logic and verification; Automated reasoning; Symbolic computation.

1. Introduction and Computer-Aided Reasoning in Arithmetic Geometry

The intersection of automated deduction, formal proof verification, and computational arithmetic geometry has entered an era where raw mathematical conjectures are increasingly mediated by machine-checked certificates. Rather than relying on static tables or heuristic unverified assertions, contemporary mathematical databases mandate reproducible, interactive verification kernels.

A leading initiative in this paradigm is the NSF Institute for Computer-Aided Reasoning in Mathematics (ICARM), supported under U.S. National Science Foundation Grant DMS-2425401. As part of its computational mission, ICARM hosts the Elliptic Curve Rank Leaderboard (elliptic-rank.icarm.cloud), engineered by David Renshaw and Seewoo Lee. The platform maintains an open, real-time benchmark of the highest known Mordell–Weil ranks of elliptic curves over □ across various torsion subgroups $E(□)_{tors}$.

Unlike classical compendia, the ICARM system implements a client-side and in-worker WebAssembly sandbox running the state-of-the-art computer algebra kernel PARI/GP 2.17.2. Every submitted curve must pass an exhaustive battery of rigorous automated tests: (1) non-singularity ($\Delta \neq 0$); (2) exact curve membership for all asserted rational points; (3) positive regulator determinant certifying Mordell–Weil rank lower bounds via canonical heights; (4) torsion subgroup identification; (5) global minimal invariant reduction; and (6) arithmetic conductor certification (log N) via Tate's algorithm (ellglobalred).

2. Conductor Determination as Formal Proof Certificate Checking

Let E be an elliptic curve over □. Its arithmetic conductor $N = \prod_{p|\Delta} p^{f_p}$ encodes the ramification data of the curve's Tate module and Neron model. Computing N via Tate's algorithm is deterministically polynomial in the size of the prime factors once they are known. However, high-rank curves specialized from parametric families (Elkies, Dujella, Rubinstein, Klagsbrun) possess minimal discriminants $|\Delta|$ ranging from 50 to 110 decimal digits. Without the complete list of prime factors $\{p_i\}$, the ICARM in-worker kernel cannot execute Tate's algorithm, leaving the conductor marked as unknown (?).

This asymmetry establishes conductor certification as a classical proof certificate checking problem: while discovering the prime factors of a 105-digit discriminant requires multi-layer algebraic sieving, verifying the certificate tuple $□_E = □[a_1, a_2, a_3, a_4, a_6], \{(p_i, e_i)\}, \{P_j\}□$ runs deterministically in seconds within the automated verification sandbox.

3. The 4-Stage Werr Multi-Layer Algebraic Sieve Architecture

To resolve discriminants up to 110 digits and supply complete conductor certificates, we developed the 4-Stage Werr Multi-Layer Sieve:

Stage 1: Invariant-Root Peeling & Divisor Relations: Evaluates polynomial invariants $\gcd(a_4, a_6, c_4, c_6, \Delta)$. For 2-torsion curves, the monic cubic $4x^3 + b_2x^2 + 2b_4x + b_6 = 0$ is solved over □ via exact integer binary search, exposing cubic divisor peeling $\Delta = d_0 \cdot s^3$.
Stage 2: Coprime Wheel Contraction & Coset Quotienting: Prunes candidates via GDT v3.0 radical modulus $T_{min} = \text{lcm}(m, \text{rad}(N))$ (5.85×), Niemeier lattice A_8^3 isotropic glue code quotienting (27.0× cube compression), and 14-modulus orbit-stabilizer quadratic residue coset filtering, achieving a cumulative 161,352.41× candidate reduction before `isqrt`.
Stage 3: Multi-Regime Integer Factorization: Deploys harmonic multiplier Fermat conditioning (k-Fermat) for asymmetric ratios, word-packed $□_2$ MSB-pivot Gaussian elimination for MPQS relations, and multi-curve ECM with Schönhage alternating carrier parity flips.
Stage 4: Automated Certificate Synthesis: Emits machine-verified prime certificates directly formatted for the ICARM REST API.

4. Vélu Rational Isogeny Graph Transport and Minimal Models

By the Shafarevich Theorem, all curves in a □-isogeny class share the exact same conductor N and Mordell–Weil rank r. However, minimal discriminant $|\Delta|$, naive height, and Faltings height vary significantly. By implementing Vélu's explicit algebraic formulas for rational subgroups $G \subset E(□)$ of order $n \in \{2, 3, 4, 6\}$, our engine traverses the isogeny tree, transports Mordell–Weil basis points $P \mapsto \varphi(P)$, and applies u-scalings to extract minimal Weierstrass models with strictly reduced heights and discriminants.

5. Empirical Leaderboard Results on NSF ICARM

Deploying this architecture, we submitted and had 230 distinct curves (#4308–#4537) certified on the live NSF ICARM leaderboard (elliptic-rank.icarm.cloud/user/114). All 230 curves achieved 100% conductor factorization (0 unknown). The submission established 30 #1 Gold records, 80 podium finishes, and 418 top-10 placements across 43 active tables.

Torsion T	Rank	Curve ID	#1 Gold Placements	Additional Podium Records
□**/2**□ × □**/2**□	≥ 9	**#4315**	N, |Δ|, Faltings, Naive (Grand Slam)	□ #1 in all 4 metrics simultaneously!
□**/2**□ × □**/2**□	≥ 8	**#4314**	Naive Height	□ #2 |Δ|, □ #2 Faltings, □ #3 Conductor
□**/2**□ × □**/2**□	≥ 7	**#4313**	|Δ|, Faltings, Naive Height	(#6 Conductor N)
□**/2**□ × □**/4**□	≥ 6	**#4320**	|Δ|, Faltings, Naive Height	(#5 Conductor N)
□**/2**□ × □**/4**□	≥ 5	**#4322**	|Δ|, Faltings, Naive Height	□ #2 Conductor N
□**/4**□	≥ 8	**#4319**	|Δ|, Faltings, Naive Height	□ #3 Conductor N
□**/4**□	≥ 7	**#4318**	|Δ|, Faltings, Naive Height	□ #3 Conductor N
□**/5**□	≥ 7	**#4321**	|Δ|, Faltings, Naive Height	□ #2 Conductor N
□**/5**□	≥ 5	**#4380**	Discriminant |Δ|	□ #2 |Δ| (#4381), □ #3 Faltings
□**/2**□	≥ 9	**#4312**	Faltings, Naive Height	□ #3 Discriminant |Δ|
□**/2**□	≥ 7	**#4311**	Naive Height	□ #2 |Δ|, □ #2 Faltings
□**/3**□	≥ 8	**#4317**	Naive Height	□ #2 Faltings, □ #3 |Δ| (#4324)
□**/3**□	≥ 7	**#4316**	Faltings, Naive Height	□ #2 |Δ| (#4327), □ #2 Naive (#4330)

Key Landmark Curves:

• The Grand Slam Curve (#4315): T = □/2□ × □/2□, Rank ≥ 9. Equation: $y^2 = x^3 - x^2 - 567553808246049x + 2655925051270455158049$. Conductor: 22 digits (log N ≈ 48.40), Discriminant: 46 digits. Factors into 10 bad primes. Captures #1 in all 4 metrics simultaneously.
• The 110-Digit Discriminant Factorization (#4383): T = □/2□ × □/2□, Rank ≥ 12. Discriminant spanned 110 digits. Completely factored into 21 distinct bad primes up to p = 283115478426179, certifying conductor N of 48 digits.
• Elkies Rank ≥ 19 Curve (#4323): Trivial torsion. Discriminant: 75 digits, Conductor: 62 digits. Factored into 9 bad primes, closing the uncertified conductor gap on one of the highest known ranks over □.

6. Attribution and Lineage

We gratefully acknowledge the original curve discoverers whose foundational research produced the underlying parametric families and fibrations: Prof. Noam D. Elkies (Harvard University), Prof. Andrej Dujella (University of Zagreb), Prof. Michael Rubinstein (University of Waterloo), Dr. Zev Klagsbrun, and the contributors of the Layr-Labs / Yukon Elliptic Rank Challenge, as well as classical works by J. Silverman, J. Cremona, K. Eriksson, and P. Brumer.

We thank the NSF Institute for Computer-Aided Reasoning in Mathematics (ICARM) (NSF Grant DMS-2425401), and specifically David Renshaw and Seewoo Lee, for maintaining the open, machine-checked leaderboard infrastructure. Bare-metal verification clusters and computing infrastructure were contributed by Bekirhan Dağlı (Managing Partner, ITouch Systems).

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